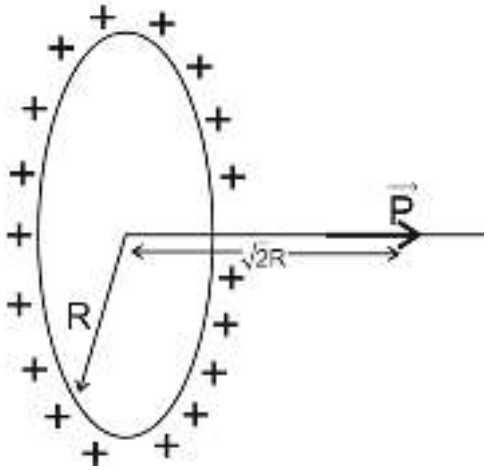
PART-1 : PHYSICS

SECTION-I

1) The net force experienced by dipole (placed along the axis of ring at distance $\sqrt{2}$ R from centre of the ring of Radius R) due to uniformly charged ring as shown in figure is



(A) Towards left

(B) Towards right

(C) Upward

(D) No force

2) **Assertion (A) :-** The electrostatic force between the plates of a charged isolated capacitor decreases when dielectric just fill the whole space between the plates. **Reason (R) :-** The electric field between the plates of a charged isolated capacitor decreases when dielectric fills the whole space.

(A) Both A and R are true and R is correct explanation of A

(B) Both A and R are true but R is not correct explanation of A

(C) Assertion A is true but Reason R is false

(D) Assertion A is false but Reason R is true

3) The electron of hydrogen atom is considered to be revolving around the fixed proton in circular orbit of radius $\frac{b^2}{me^2}$ with velocity $\frac{e^2}{b}$ where $b = \frac{h}{2\pi}$, the electric current associated due to this revolution is

(A) $\frac{2\pi^2 me^5}{h^3}$

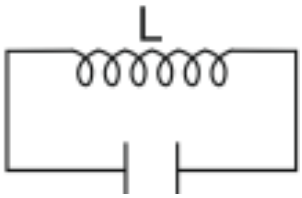
(B) $\frac{4\pi^2 me^5}{h^3}$

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(C) $\frac{2\pi me^5}{h^3}$

(D) $\frac{4\pi me^5}{h^3}$

4) At time t = 0, the LC circuit shown in the figure has equal amount of energy stored in the capacitor C = 20 μF and in the inductor, each of 200μJ. Current amplitude is $\frac{1}{\sqrt{10}}$A. The value of inductor is



(A) 2 mH

(B) 4 mH

(C) 8 mH

(D) 16 mH

5) A negatively charged ion (with charge –q and mass m) moving with velocity $v_0\hat{i}$ enters a region of uniform magnetic field of Intensity $B_0\hat{k}$ at time t = 0. Here B_0 and v_0 are positive constants. The angular momentum of ion at time $t = \frac{\pi m}{2B_0q}$ about the point where the ion enters the region of field is

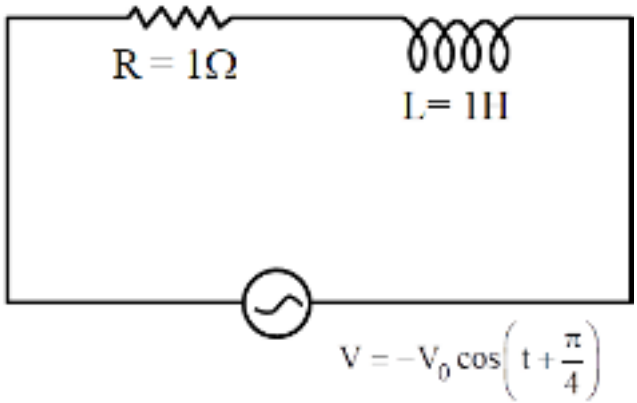
(A) $\frac{m^2v_0^2}{B_0q}\hat{k}$

(B) $\frac{m^2v_0^2}{B_0q}\left(-\hat{k}\right)$

(C) zero

(D) $\frac{m^2v_0^2}{2B_0q}\left(\hat{k}\right)$

6) The instantaneous voltage drop V_R across resistor in circuit is



(A) $V_R = \frac{-V_0}{\sqrt{2}}\cos t$

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(B) $V_R = V_0 \cos t$

(C) $V_R = \frac{-V_0}{\sqrt{2}} \cos\left(t + \frac{\pi}{4}\right)$

(D) $V_R = \frac{-V_0}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)$

7) A parallel plate capacitor (with circular plates of radius r and plate separation d) is being charged. Magnitude of electric field in between the plates of capacitor at any time t is E. Magnitude of poynting vector at outer edge of capacitor plate just inside the plate is (where μ_0 and ϵ_0 respectively represent magnetic permeability and electrical permittivity of free space)

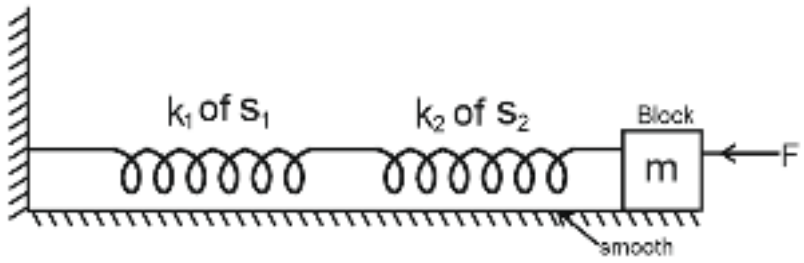
(A) $\frac{1}{2}\epsilon_0 rE\left(\frac{dE}{dt}\right)$

(B) $\frac{1}{2}\mu_0 rE\left(\frac{dE}{dt}\right)$

(C) $\frac{1}{2}\frac{\epsilon_0}{\mu_0} rE\left(\frac{dE}{dt}\right)$

(D) $\frac{1}{2}\frac{\mu_0}{\epsilon_0} rE\left(\frac{dE}{dt}\right)$

8) In the given figure there are two ideal, relaxed springs S_1 & S_2 of spring constants k_1 & k_2 respectively. An application of external Variable force F causes block to move very slowly towards the wall on smooth horizontal floor by a distance x.



	Column-I		Column-II
(I)	Work done by S_2 on block	(P)	Zero
(II)	Work done by S_2 on S_1	(Q)	$-\frac{1}{2}\left(\frac{k_1k_2}{k_1+k_2}\right)x^2$
(III)	Work done by F on block	(R)	$\frac{1}{2}\left(\frac{k_1k_2}{k_1+k_2}\right)x^2$
(IV)	Work done by S_1 on wall	(S)	$\frac{1}{2}\frac{k_1k_2^2}{(k_1+k_2)^2}x^2$

(A) I – Q ; II – S ; III – R ; IV – P

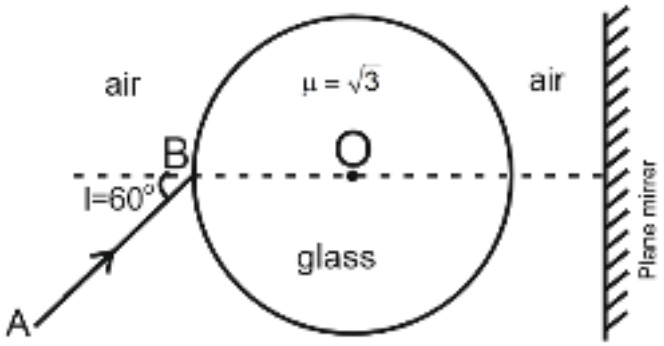
(B) I – R ; II – S ; III – Q ; IV – P

(C) I – S ; II – R ; III – P ; IV – Q

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(D) I – R ; II – S ; III – P ; IV – Q

9) A light ray AB is incident on solid sphere of glass of refractive index $\sqrt{3}$. A mirror is placed at some distance which is perpendicular to BO. Where O is centre of sphere and B is incident point of ray, then final emergent ray from glass sphere after reflection from plane mirror is :-



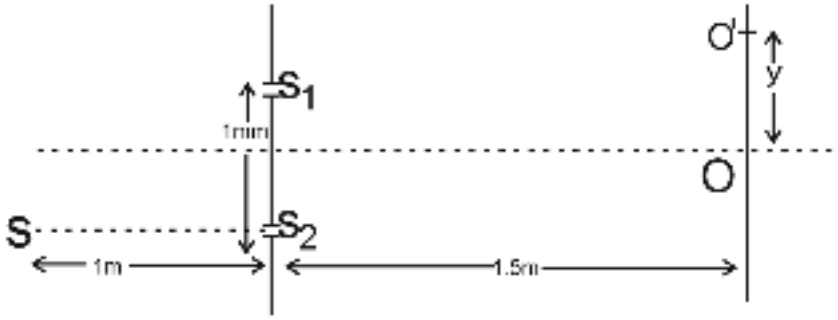
(A) Horizontal and parallel to BO

(B) Vertical and perpendicular to BO

(C) Ray will retrace it's path and finally emerge out from point B as BA

(D) Light can not emerge out from sphere as T.I.R takes place from sphere to air boundary

10) In the following YDSE setup, there is monochromatic light source S of wavelength 5000 Å is placed 1m behind the plane of slits and in front of one of the slits, then central maxima O' is shifted by distance y from central point O as shown in figure then



(A) y = 1 mm and Ist maxima at O

(B) y = 1 mm and Ist minima at O

(C) y = 0.75 mm and Ist minima at O

(D) y = 0.75 mm and Ist maxima at O

11) If E_n represents the kinetic energy of electron in nth orbit of hydrogen like atom then select the correct relation from the following.

(A) $E_n = \frac{E_{n+1} + E_{n-1}}{2}$

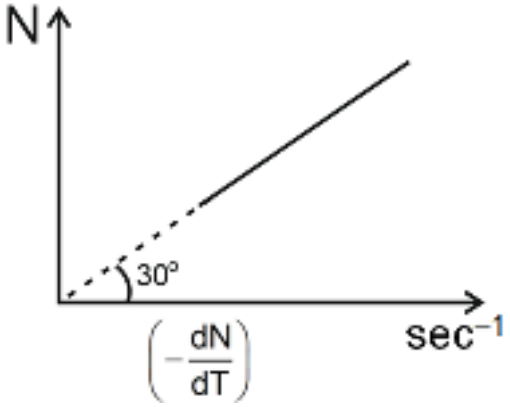
(B) $E_n < \frac{E_{n+1} + E_{n-1}}{2}$

(C) $E_n > \frac{E_{n+1} + E_{n-1}}{2}$

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(D) $E_n = \frac{E_{n+1} - E_{n-1}}{2}$

12) A graph is plotted between rate of disintegration $\left(\frac{-dN}{dt}\right)$ and number of active nuclei at any instant. What is half life of radio active substance? (If time in seconds)



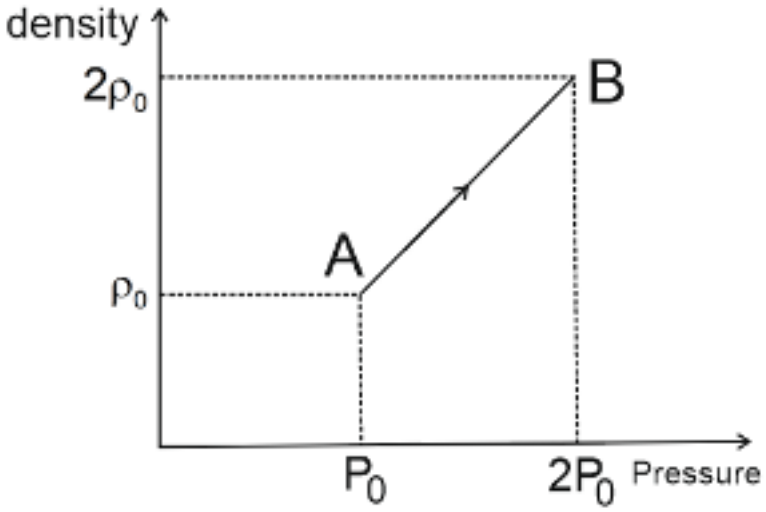
(A) $\sqrt{3}$ sec

(B) $\frac{1}{\sqrt{3}}$ sec

(C) $\sqrt{3}\ell n2$ sec

(D) $\frac{\ell n2}{\sqrt{3}}$ sec

13) The density versus pressure graph of one mole of an ideal monoatomic gas of molar mass M undergoing a process AB is shown in figure. The amount of heat rejected by gas in the process is



(A) $\frac{2P_0M}{\rho_0}\ell n2$

(B) $\frac{P_0M}{\rho_0}\ell n2$

(C) $\frac{1}{2}\frac{P_0M}{\rho_0}\ell n2$

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(D) $4\frac{P_0M}{\rho_0}\ell n2$

14) **Statement-I :-** Two particles of mass 1 kg and 3 kg move towards each other under their mutual force of attraction. No other force acts on them. When the relative velocity of approach of two particle is 2 m/sec, their centre of mass has velocity of 0.5 m/sec. When the relative velocity of approach increases to 3 m/sec, the velocity of the centre of mass decreases to 0.25 m/sec

Statement-II :- The total kinetic energy as seen from ground is $\frac{1}{2}\mu V_{rel}^2 + \frac{1}{2}MV_{c.m}^2$ and here μ is reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ and M is total mass M = (m_1 + m_2) and in absence of external force, total mechanical energy remains conserved

(A) Statement-I is true, Statement-II is true

(B) Statement-I is false, Statement-II is true

(C) Statement-I is false, Statement-II is false

(D) Statement-I is true, Statement-II is false

15) A particle of mass m is moving in a horizontal circle of Radius R under a centripetal force given by $\left(F_{cen} = \frac{-2}{R^2}\right)$ then. (If potential energy at infinity is zero)

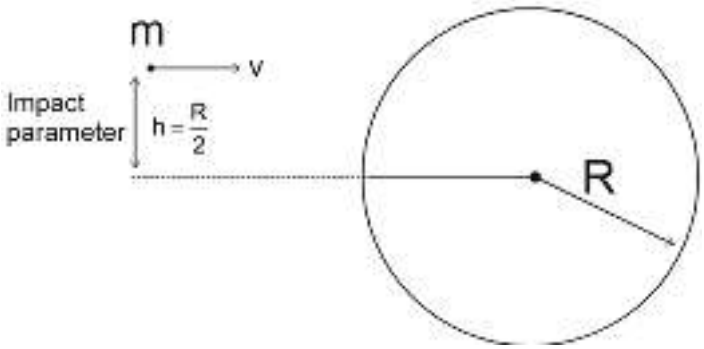
(A) Kinetic energy is $\frac{1}{6R}$

(B) Total energy of particle is $\frac{-1}{R}$

(C) Kinetic energy of particle is $\frac{2}{R}$

(D) Potential energy is $\frac{1}{R}$

16) A particle of mass m strikes a smooth disc of radius R and of same mass m with velocity v. Initially the disc is at rest. The coefficient of restitution e = 0.5. Find the speed of particle just after the impact. Assuming the disc is placed on smooth horizontal table as shown by top view of table



(A) $\frac{3v}{4}\sqrt{19}$

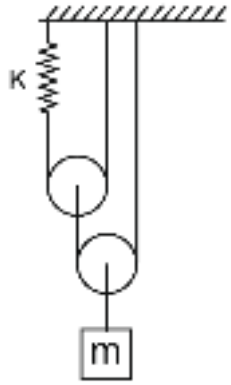
(B) $\frac{v}{4}\sqrt{19}$

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(C) $\frac{v}{2}\sqrt{19}$

(D) $\frac{v}{8}\sqrt{19}$

17) The time period of following oscillation system is :- (All pulley and spring are ideal)



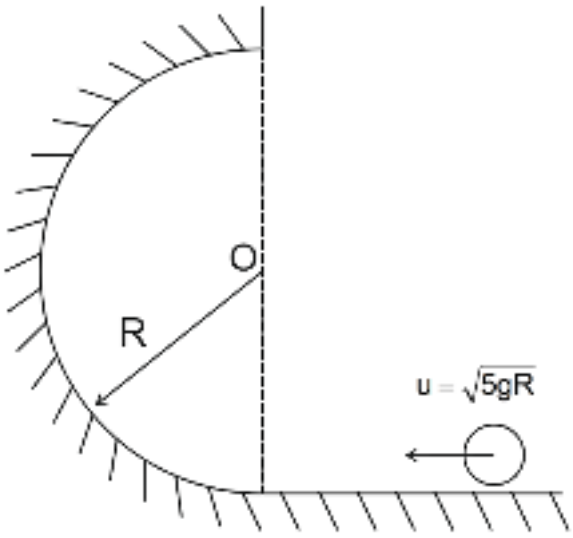
(A) $\frac{\pi}{2}\sqrt{\frac{m}{k}}$

(B) $2\pi\sqrt{\frac{m}{32k}}$

(C) $2\pi\sqrt{\frac{16m}{k}}$

(D) $4\pi\sqrt{\frac{m}{k}}$

18) A ball is projected on a smooth horizontal surface towards a smooth circular track of radius R with speed $\sqrt{5gR}$. The ball after leaving the semi-circular track lands on the smooth ground where coefficient of restitution is $\frac{1}{2}$. The range between first collision and second collision is



(A) R

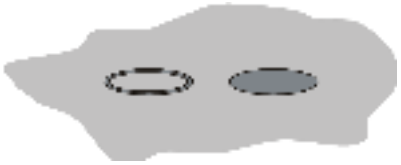
(B) $\frac{3R}{2}$

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(C) $\frac{2R}{3}$

(D) 2R

19) A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses (thickness may be different)?



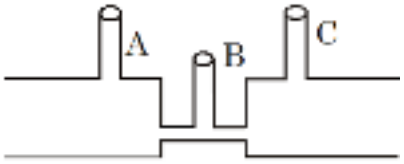
(A) Both have the same mass.

(B) Mass of the ring is half of that of the disk.

(C) Mass of the ring is double to that of the disk.

(D) More information is needed to decide.

20) Three tubes A, B, C are connected to a horizontal pipe in which liquid is flowing. The radii of the pipes at the joints of A, B and C are 2 cm, 1 cm and 2 cm respectively. The height of the liquid



(A) in A is lesser than C

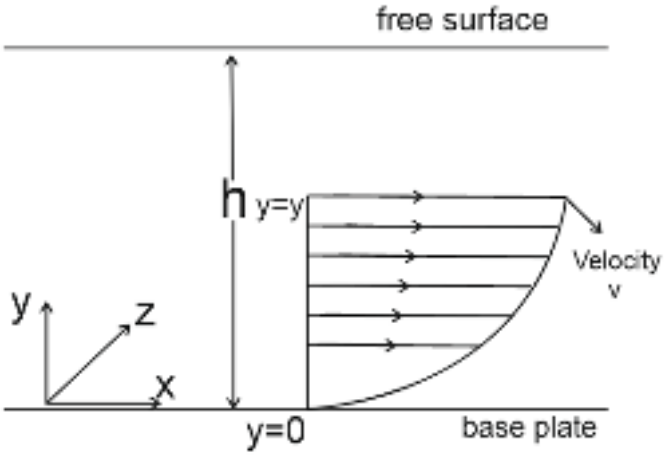
(B) in A and C is equal

(C) is same in all the three

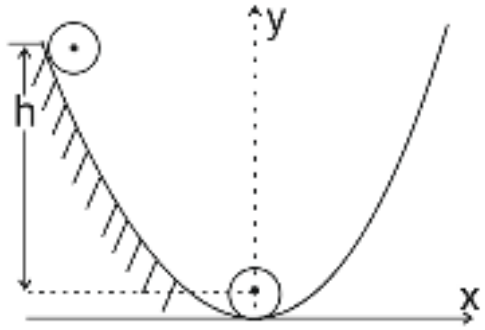
(D) in A and B is same

SECTION-II

1) For the situation shown in figure liquid of coefficient of viscosity η flown over the surface of fixed plate in x-z plane. The velocity of liquid as a function of y (perpendicular distance from plate to free surface) is given as $v = \alpha\left[\frac{y}{h} - \left(\frac{y}{h}\right)^2\right]$. If magnitude of shear stress that liquid applies at base plate is $\frac{N\alpha\eta}{h}$ then N is

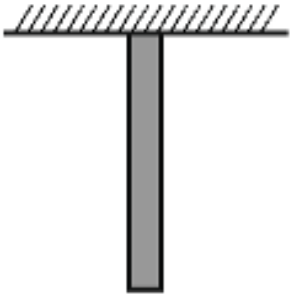


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2) 

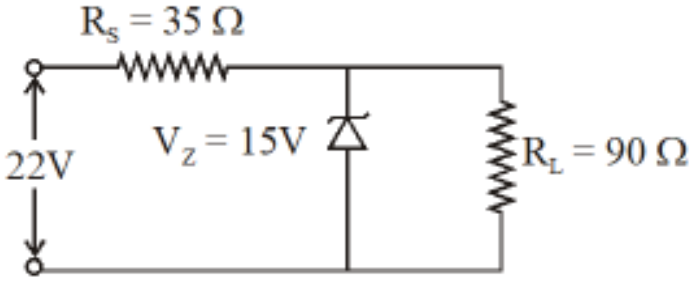
A solid sphere is released from a point at a height h = 7m on a parabolic surface as shown in figure. Left part of the surface (x< 0) is sufficiently rough to avoid slipping and right part of the surface (x>0) is perfectly smooth. The maximum height in meter upto which the sphere will reach on right part, is

3) A uniform rod of mass M, length □ and area of cross section A is hanging vertically from ceiling as shown. If Young's modulus of elasticity is Y, the elastic potential energy stored in upper half of the rod, is $\frac{14M^2g^2\ell}{nAY}$, then value of n is _____.



4) A reversible heat engine has an efficiency of 33.3%, removing heat from a hot reservoir and rejecting heat to a cold reservoir at 0°C. If the engine now operates in reverse, how long would it take of freeze 1.0 kg of water at 0°C, if it operates on a power of 186 W?

5) The value of power dissipated across the zener diode (V_z = 15 V) connected in the circuit as shown in the figure is x × 10^{-1} watt. The value of x, to the nearest integer, is _____.



PART-2 : CHEMISTRY

SECTION-I

1) Which of the following statement is correct regarding the line spectrum of Hydrogen & Hydrogen like species ?

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(A) All the lines of Balmer series for H–atom belongs to the visible region.

(B) The first line of Balmer series for H–atom will be red line.

(C) Balmer series for He^+ ion belongs to the visible region.

(D) Second line of Balmer series of H–atom is red line.

2) Which of the following Processes are exothermic ?
(1) $O^-(g) \rightarrow O^+(g)$
(2) $Be^-(g) \rightarrow Be(g)$
(3) $P(g) \rightarrow P^{3-}(g)$
(4) $H^+(g) \rightarrow H^-(g)$

(A) 1,2,4

(B) 2,4

(C) 2,3

(D) 1,3

3) **Assertion :-** Solubility of $Pb(OH)_2$ (weak base) decreases on addition of NaOH.
Reason :- NaOH increases OH^- concentration forcing the equilibrium to shift in the backward direction.

(A) Assertion and Reason both are correct and reason is correct explanation of assertion.

(B) Assertion and Reason both are correct and reason is incorrect explanation of assertion.

(C) Assertion and Reason both are incorrect.

(D) Assertion is correct and Reason is incorrect.

4) Which of the following statement is correct?

(A) As per MOT, double bond in C_2 consist of sigma bond and pi bond.

(B) Bond energy : $N_2 > C_2 > O_2 > B_2 > F_2$

(C) Bond length : $N_2 < C_2 < O_2 < B_2 < F_2$

(D) Diamagnetic Li_2 molecules are known to exist in solid phase.

5) The aqueous solution of which of the following will have highest freezing point?

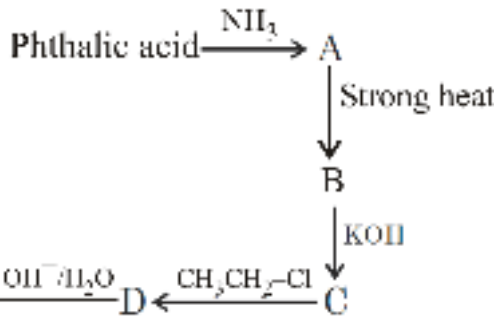
(A) Glucose (0.1 m)

(B) NaCl (0.1 m)

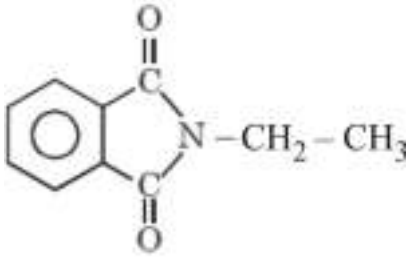
(C) Na_3PO_4 (0.05 m)

(D) CH_3COOH (0.2 m, degree of dissociation = 0.1)

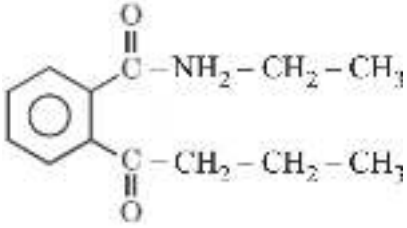
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6) The structure of E is :-

(A) 

(B) CH_3CH_2OH

(C) 

(D) $CH_3CH_2NH_2$

7) Two system $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ and $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ are simultaneously in equilibrium in a vessel at constant volume. If some CO is introduced in the vessel, then at new equilibrium, the moles of :

(A) PCl_5 increase

(B) PCl_5 remain unchanged

(C) PCl_5 decrease

(D) Cl_2 increase

8) During estimation of nitrogen present in an organic compound by kjeldahl's method, the ammonia evolved from 'm' g of the compound in kjeldahl's estimation of nitrogen, was absorbed in 'V'ml of 'M' molar H_2SO_4 solution. The excess of acid required 'V_1'ml of NaOH of molarity 'M'. The % of 'N' in the organic sample is :-

(A) $\frac{28 \times M \times 2\left(V - \frac{V_1}{2}\right)}{1000} \times \frac{100}{m}$

(B) $\frac{14 \times M \times \left(V - \frac{V_1}{2}\right)}{1000} \times \frac{100}{m}$

(C) $\frac{14 \times M \times 2(2V - V_1)}{1000} \times \frac{100}{m}$

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(D) $\frac{28 \times M\left(V-\frac{V_1}{2}\right)}{1000} \times \frac{100}{m}$

9) Match the column and choose the correct option :

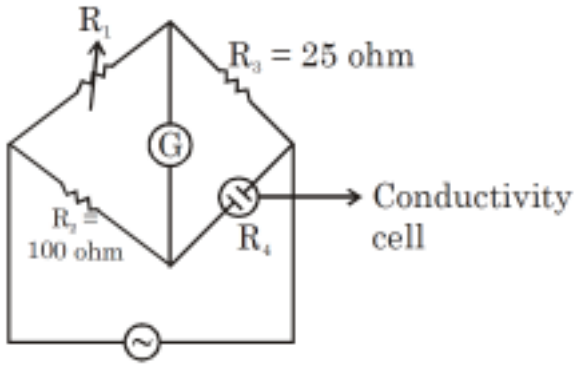
Column-I (multiple)		Column-II (Prefix)	
(A)	10^{9}	(1)	atto
(B)	10^{21}	(2)	zeta
(C)	10^{-18}	(3)	exa
(D)	10^{18}	(4)	giga

(A) A – 2 ; B – 1 ; C – 3 ; D – 4

(B) A – 4 ; B – 2 ; C – 1 ; D – 3

(C) A – 1 ; B – 2 ; C – 3 ; D – 4

(D) A – 3 ; B – 2 ; C – 4 ; D – 1



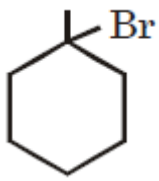
10) The galvanometer shows no deflection when R_1 = 100 ohm. Find the molar conductivity of 0.05 M KCl solution, if the cell constant is 1.25 cm^{-1}.

(A) 200 S cm^2 mol^{-1}

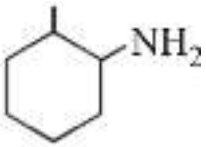
(B) 400 S cm^2 mol^{-1}

(C) 1000 S cm^2 mol^{-1}

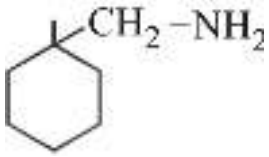
(D) 125 S cm^2 mol^{-1}

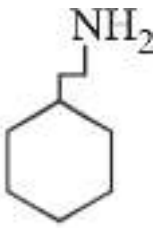
11) 

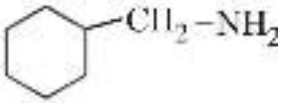
$\xrightarrow[(CH_3)_3COH]{(CH_3)_3CO^-K^+}$ A $\xrightarrow[R-O-O-R/h\nu]{HBr}$ B $\xrightarrow{AgCN}$ C $\xrightarrow{H_3O^+}$ D What is the correct structure of D is :-

(A) 

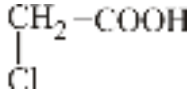
For More Material Join: @JEEAdvanced_2025

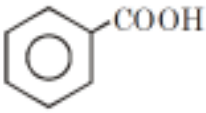
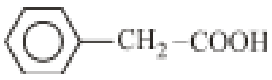
(B) 

(C) 

(D) 

12) Give the correct order of Acidic strength.

(1) HCOOH (2) 

(3)  (4) 

(A) (1) > (2) > (3) > (4)

(B) (3) > (1) > (2) > (4)

(C) (2) > (1) > (4) > (3)

(D) (2) > (1) > (3) > (4)

13) Consider the complex : $[Ni(NH_3)_{6-2n}(en)_n]^{2+}$ The maximum number of sterioisomers is possible for -

(A) n = 0

(B) n = 1

(C) n = 2

(D) n = 3

14) For the complex $[Co(C_2O_4)_3]^{3-}$, which information is "Incorrect" ?

(A) $C_2O_4^{2-}$ is monodentate ligand.

(B) Inner orbital complex.

(C) Oxidation state of Co = +3.

(D) Complex has octahedral geometry.

15) **Statement-1 :** Benzaldehyde gives positive test with both Tollens' reagent and Fehling's reagent.
Statement-2 : Ph – CH_2 – CHO gives positive test with Fehling's reagent.

(A) Statement-1 is true, statement-2 is false.

(B) Statement-1 is false, statement-2 is true.

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(C) Both Statement-1 and statement-2 are true.

(D) Both Statement-1 and statement-2 are false.

16) A compound 'X' having molecular formula $C_4H_{11}N$ give reaction with $C_6H_5SO_2Cl$ and give compounds 'Y' and HCl, But does not give reaction with chloroform in presence of KOH. Compound 'X' also undergo acylation with ethanoyl chloride to yield 'Z'.
Choose incorrect statement.

(A) Compound 'Y' is insoluble in NaOH.

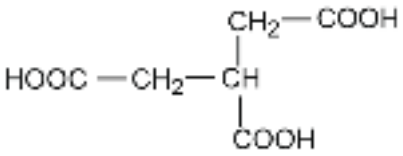
(B) 'Z' is an Amide.

(C) 'X' reacts with nitrous acid to give N_2 gas.

(D) Compound 'X' does not give mustard oil test.

17)

What is the common name of the following compound ?



(A) Tricarboxylic acid

(B) Carballylic acid

(C) Propane tricarboxylic acid

(D) Hexane trioic acid

18) What will be the expression of rate constant (k) of reaction in terms of pressure for following reaction, when the initial pressure is P_0 and, at time 't', total pressure is P_T (considering reaction is following first order kinetics) : $2A(g) \rightarrow B(g)$

(A) $k = \frac{1}{t}\ell n\frac{P_0}{2P_0 - P_T}$

(B) $k = \frac{1}{2t}\ell n\frac{P_0}{P_0 - P_T}$

(C) $k = \frac{1}{2t}\ell n\frac{P_0}{2P_T - P_0}$

(D) $k = \frac{2.303}{t}\ell og\frac{P_0}{P_T}$

19) Calculate enthalpy of reaction for following reaction -
$C_3H_8(g) + H_2(g) \rightarrow CH_4(g) + C_2H_6(g)$
Given that -
Enthalpy of combustions :
Δ_cH^o (Graphite, s) = –393 kJ mol^{-1}
Δ_cH^o (CH_4, g) = –890 $kJmol^{-1}$
Δ_cH^o (C_2H_6, g) = –1560 $kJmol^{-1}$
and enthalpy of formation :
Δ_fH^o (Propane, g) = –104 $kJmol^{-1}$

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$\Delta_f H^\circ$ (H_2O, □) = –285 kJmol^{-1}

(A) –50 kJmol^{-1}

(B) +2061 kJmol^{-1}

(C) –2061 kJmol^{-1}

(D) +50 kJmol^{-1}

20) What is the f-subshell electronic configuration of Am^{3+} (Z = 95)?

(A) $5f^4$

(B) $5f^5$

(C) $5f^6$

(D) $5f^7$

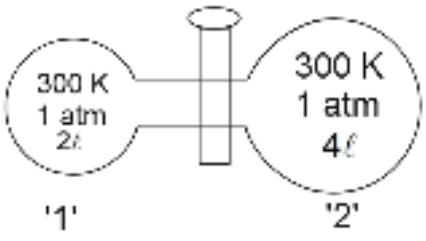
SECTION-II

1) According to VSEPR theory, how many single super pairs are present in $C_2(CN)_4$

2) Iron (III) chloride (aq.) is treated with potassium ferrocyanide to give a blue ppt 'A'. Which on further treatment with excess of potassium ferrocyanide give a colloidal solution 'B'.
Find the sum of number of Iron atoms present in 'A' and 'B'.

3) Find the molecular mass of Tripeptide formed using Amino acids – Glycine, Alanine, Leucine.

4) How many of the following compound will give turbidity in 5 to 10 minutes with Lucas Reagent ?
Methyl Alcohol, n-Propyl alcohol, Isopropyl alcohol, n-Butyl alcohol, sec-Butyl alcohol, Isobutyl alcohol, tert-Butyl alcohol.



5)
Two flasks '1' & '2' are filled with H_2 gas as shown above connected by a valve of negligible volume. When the valve is opened, what will be the change in entropy, (in cal K^{-1}) due to mixing.
Given : ln 2 = 0.7 R = 2 cal mol^{-1} K^{-1}

PART-3 : MATHEMATICS

SECTION-I

For More Material Join: @JEEAdvanced_2025

1) Sum of series $2+\frac{5}{7}+\frac{8}{49}+\frac{11}{343}+....\infty$ is

(A) $\frac{35}{12}$

(B) $\frac{49}{18}$

(C) $\frac{35}{16}$

(D) $\frac{7}{12}$

2) Given a right circular cone has a volume V_1 and the largest right circular cylinder that can be inscribed in the cone has a volume V_2 . Then $V_1 : V_2$ is $\frac{a}{b}$ (where a, b are coprime). Find the value of $a^2 - b^2$

(A) 25

(B) 65

(C) 64

(D) 20

3) **Assertion (A) :** Line $7x + 5y + 1 = 0$ cannot intersect the parabola $y = x^2$ in a rational point (whose abscissa & ordinate both are rational)
Reason (R) : For odd integer a,b and c equation $ax^2 + bx + c = 0$ cannot have rational roots.

(A) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(B) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

(C) **A** is correct but **R** is not correct

(D) **A** is not correct but **R** is correct

4) Let $f : (1, \infty) \to (0, \infty)$ be a continuous decreasing function with $\lim_{x\to\infty} \frac{f(x)}{f(8x)} = 1$ Then $\lim_{x\to\infty} \frac{f(5x)}{f(8x)}$ is equal to

(A) $\frac{4}{8}$

(B) $\frac{4}{6}$

(C) $\frac{6}{8}$

(D) 1

5) Let a ray of light be travelling along the, line : $4x + 3y - 2k = 0$ and strike, a parabolic mirror at $(1, p)$. After reflection it moves, along the : line $3x - 4y + k = 0$. If the directrix of the parabola is the line $3x - 4y + 15 = 0$ then the length of latus-rectum of the parabola is :

(A) 4

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(B) 6

(C) 8

(D) 13

6)

If AB is a chord of the circle $x^2 + y^2 = 25$. The tangents at A and B intersect at C. If (2,3) is the mid point of AB, then the area of the quadrilateral OACB is, where O is the origin.

(A) $\sqrt{\frac{3}{13}}$ sq. unit

(B) $50\sqrt{\frac{3}{13}}$ sq. unit

(C) 100 sq. unit

(D) $100\sqrt{3}$ sq. unit

7) A company has 10 employees. The company has decided to form a team including at least three employees and also excluding at least three employees. Then the number of ways of forming the team is

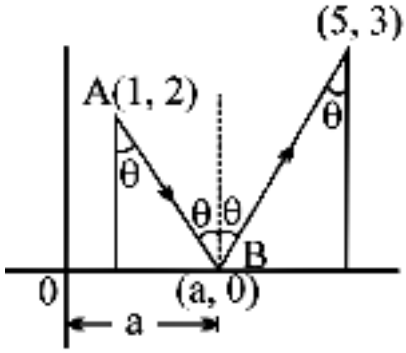
(A) 950

(B) 912

(C) 1000

(D) 953

8) A ray of light passing through the point A(1, 2) is reflected at a point B on the x – axis and then passes through (5, 3). Then the equation of AB is :



(A) 5x + 4y = 13

(B) 5x – 4y = –3

(C) 4x + 5y = 14

(D) 4x – 5y = –6

9)

List-I	List-II

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(A)	If α,β (α < β) are the two roots of the equation $\frac{1-8(\log_{10}x)^2}{\log_{10}x-2(\log_{10}x)^2}=1$ the the value of $\frac{(\beta^3+1)}{\alpha^2\beta^3}$ is equal to	(I)	5
(B)	Number of solution(s) of equation $\log_2(x^2+3)=\frac{1}{2}\log_{1/3}\left(x+\frac{1}{x}\right), x>0$ is	(II)	11
(C)	Integers satisfying the equation $\lvert x\rvert+\left\lvert\frac{4-x^2}{x}\right\rvert=\left\lvert\frac{4}{x}\right\rvert$ is/are	(III)	0
(D)	If $\log_c 2\ \log_b 125 = \log_{10} 8\log_c 10$ where c > 0, c ≠ 1, b > 0, b ≠1, then b is	(IV)	1

(A) A → I, B → III, C → IV, D → II

(B) A → II, B → III, C → IV, D → I

(C) A → IV, B → II, C → IV, D → I

(D) A → IV, B → III, C → II, D → I

10) The A.M. of n observations is M. If the sum of (n – 4) observation is a, then the mean of remaining 4 observations is (a ≠ nM) :

(A) $\frac{nM-a}{4}$

(B) $\frac{nM+a}{4}$

(C) $\frac{nM-a}{2}$

(D) nM + a

11) If $\int\frac{(\sqrt{x})^5}{(\sqrt{x})^7+x^6}dx=\frac{1}{k}\ln\left(\frac{x^k}{1+x^k}\right)+c$ then k is equal to :
(where c is constant of integration)

(A) $\frac{1}{5}$

(B) $\frac{2}{5}$

(C) 5

(D) $\frac{5}{2}$

12) Number of roots of the function f (x) = $\frac{1}{(x+1)^3}$ – 3x + sin x is :

(A) 0

(B) 1

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(C) 2

(D) More than 2

13) The value of the expression, $14\tan\left(\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}+\tan^{-1}\frac{1}{21}+\tan^{-1}\frac{1}{31}\right)$ is an integer which is equal to :

(A) 2

(B) 5

(C) 7

(D) 10

14) $\lim\limits_{x\to 0}\frac{(e^{2x^2}-1-2x^2)(\cos x-1)}{(\sin 3x-\ln(1+3x))\,x^4}$:

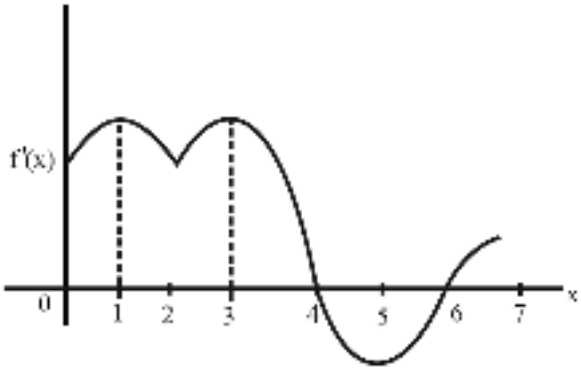
(A) $\frac{2}{9}$

(B) $\frac{1}{9}$

(C) $-\frac{1}{9}$

(D) $-\frac{2}{9}$

15) Let f(x) be a continous function in (0, 7) and graph of $f'(x)$ is given below



S_1 : f(x) has three points of extrema

S_2 : f(x) has 4 points of inflection

(A) Both statement-I and statement-II are correct.

(B) Both statement-I and statement-II are incorrect.

(C) Statement-I is incorrect and statement-II is correct.

(D) Statement-I is correct and statement-II is incorrect.

16) If the ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{\alpha}=\frac{1}{25}$ are orthogonal, then find the value of $4e_1$. (where e_1 is the eccentricity of the hyperbola)

(A) 10

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(B) 5

(C) 8

(D) 6

17) If $f(x) = 2^x + x^3 + 1$ and $g(x) = f^{-1}(x)$ then value of $\frac{g'(2)}{g''(2)}$ lies in

(A) Set of irrational numbers

(B) Set of prime numbers

(C) $[-10, 10]$

(D) None of these

18) $\lim\limits_{t\to 0} \frac{12^t - 3^t - 4^t + 1}{\sqrt{2\cos t + 7} - 3}$ is equal to

(A) $\log_e 12$

(B) $\log_e 3.\log_e 4$

(C) $(\log_e 3.\log_e 4)^{-1}$

(D) $-6\log_e 3.\log_e 4$

19) Let $\vec{a}, \vec{b}$ be two non collinear unit vectors and $\vec{x} = \vec{a} - (\vec{a}.\vec{b})\vec{b}$ and $\vec{y} = \vec{a} \times \vec{b}$. Now consider the following statements : **Statement-I :** $|\vec{x}| = |\vec{y}|$

Statement-II : $|\vec{y}| = |\vec{x}| + |\vec{x}.\vec{b}|$

(A) Statement-I is true but Statement-II is false

(B) Both Statement-I and Statement-II are false

(C) Both Statement-I and Statement-II are true

(D) Statement-I is false but Statement-II is true

20) If $I_1 = \int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}} dx$ & $I_2 = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + \pi^x} dx$, then -

(A) $\frac{1}{I_1} - \frac{1}{I_2} = \frac{3}{\pi}$

(B) $\frac{2}{I_1} - \frac{1}{I_2} = \frac{5}{\pi}$

(C) $\frac{2}{I_1} + \frac{1}{I_2} = \frac{8}{\pi}$

(D) $\frac{1}{I_1} + \frac{1}{I_2} = \frac{6}{\pi}$

SECTION-II

For More Material Join: @JEEAdvanced_2025

1) The probability of getting either all heads or all tails for exactly the second time in the 3rd trial, if in each trial three coins are tossed, is p then the value of (64p) equals to -

2) The value of $\int_{-3}^{-1} (x+1)^4(x+2)^3(x+3)^4 dx$ is

3) Number of roots of the equation $\cos^2 x + \frac{\left(\sqrt{3}+1\right)}{2}\sin x - \frac{\sqrt{3}}{4} - 1 = 0$ which lie in the interval [–π, π] is

4) The smallest possible natural number 'p' for which the equation $x^2 - px + 1258 = 0$ has integral roots ?

5) If x, y, z be three distinct real numbers, and $x \neq 1,\ y \neq 1,\ z \neq 1$. If the vectors $\vec{p} = x\hat{i} + \hat{j} + \hat{k}$, $\vec{q} = \hat{i} + y\hat{j} + \hat{k}$, $\vec{r} = \hat{i} + \hat{j} + z\hat{k}$ are such that $(\vec{p} \times \vec{q}) \cdot \vec{r} = 0$, then $\frac{1}{1-x} + \frac{1}{1-y} + \frac{1}{1-z}$ is equal to

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	D	B	C	A	A	A	A	C	D	B	D	B	B	B	D	A	D	C	B

SECTION-II

Q.	21	22	23	24	25
A.	1	5	96	15	5

PART-2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	B	B	C	B	A	D	C	D	B	C	D	D	C	A	B	C	B	C	A	C

SECTION-II

Q.	46	47	48	49	50
A.	5	9	259	2	0

PART-3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	A	B	A	D	A	B	B	A	B	A	D	C	D	D	C	B	C	D	C	D

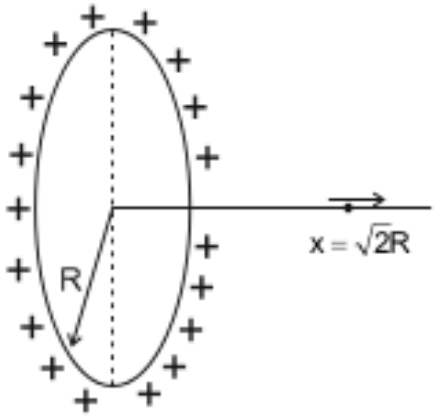
SECTION-II

Q.	71	72	73	74	75
A.	6	0	4	71	1

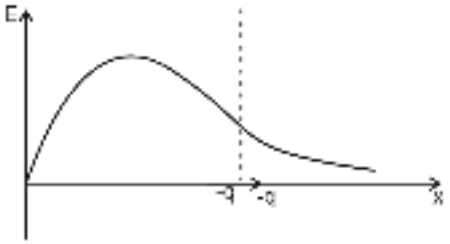
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SOLUTIONS

PART-1 : PHYSICS



1)



$\vec{F}_{dipole} = \vec{F}_{-q} + \vec{F}_{+q}$

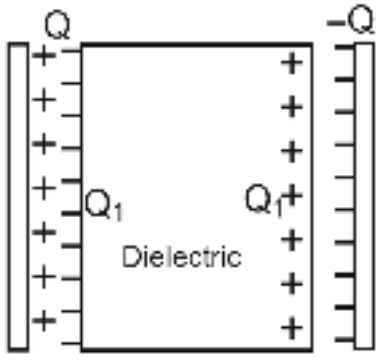
= –q(E + dE)(Leftward)+q(E – dE)(Rightward)

= Leftward

Or

$F_{Dipole} = P\frac{dE}{dx}, \frac{dE}{dx}$ is – ive at $x = \sqrt{2}R$

= Leftward



2) For isolated system, charge is constant & force between plates is the force experienced by one plate due to field of other

$\vec{F}_{-Q}$ = –Q (Electric field at negative charged plate)

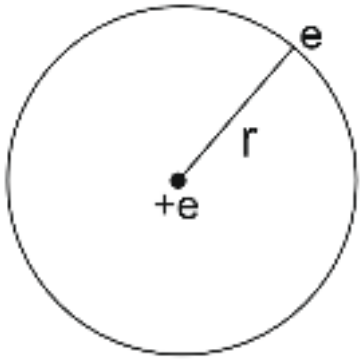
$= -Q\left(\vec{E}_{+} + \vec{E}_{dielectric}\right)$

$\vec{E}_{dielectric}$ is zero,

$\vec{E}_{+}$ is electric field due to positive plate, which remains unchanged even after introduction of

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dielectric



3)

$r = \frac{b^2}{me^2}, v = \frac{e^2}{b}$

Angular momentum = mvr = $n\frac{h}{2\pi}$

$= m \times \frac{e^2}{b} \times \frac{b^2}{me^2} = b = n\frac{h}{2\pi}$

$n = 1 \Rightarrow b = \frac{h}{2\pi}$

Electric current associated with revolution = $\frac{e}{T}$

T is time period = $\frac{2\pi r}{v} \Rightarrow i = \frac{ev}{2\pi r} = \frac{e \times e^2/b}{2\pi \frac{b^2}{me^2}}$

$i = \frac{me^5}{2\pi b^3} = \frac{me^5}{2\pi\left(\frac{h}{2\pi}\right)^3} = \frac{4\pi^2 me^5}{h^3}$

4)

In LC circuit $q = q_0 \sin(\omega t+\varphi)$

$i = q\omega \cos(\omega t + \phi)$

$i_0 = q_0\omega, \quad \omega = \frac{1}{\sqrt{LC}}$

From energy conservation,

$E_{Total} = E_{mag.} + E_{electrostatic}$

$\frac{1}{2}Li_0^2 = \frac{q_0^2}{2C} = \frac{1}{2}Li^2 + \frac{q^2}{2C} = 2\frac{q^2}{2C}$

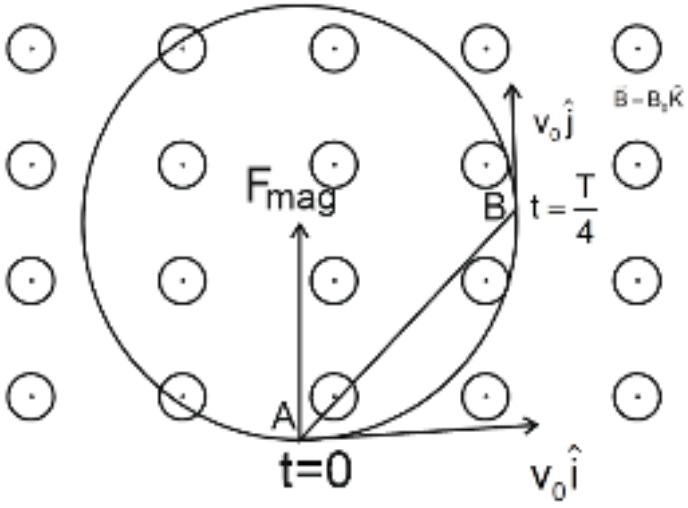
$q = \frac{q_0}{\sqrt{2}} \Rightarrow \sin(\omega t + \phi) = \frac{\pi}{4}$

$t = 0 \Rightarrow E_{Capacitor} = E = \frac{q^2}{2C} = \frac{\left(q_0/\sqrt{2}\right)^2}{2C}$

$q_0 = \sqrt{4 \times 200 \times 20 \times 10^{-12}} = \frac{4 \times 10^{-4}}{\sqrt{10}}C$

L = 8mH

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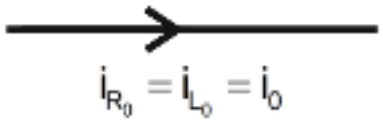
5)

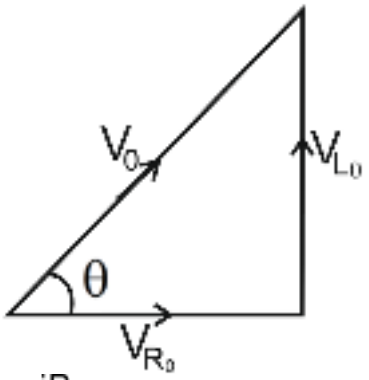
$$t=\frac{\pi m}{2B_0q}=\frac{T}{4}$$

$\vec{L}$ = Angular momentum about point A

$$=m\vec{r}_{B/A}\times\vec{v}$$
$$=mR(i+j)\times v_0 j$$
$$=mR(i+j)\times v_0 j=mv_0R\hat{K}$$
$$=mv_0\frac{mv_0}{B_0q}\hat{K}=\frac{m^2{v_0}^2}{B_0q}\hat{K}$$

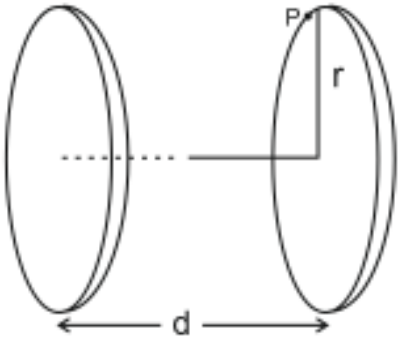
6)





$$V_R=iR$$
$$Z=\sqrt{R^2+{X_L}^2}=\sqrt{1^2+1^2}=\sqrt{2}$$
$$\omega=1 \qquad \tan\phi=\frac{X_L}{R}=1\Rightarrow\phi=\frac{\pi}{4}$$
$$V=-V_0\cos\left(t+\frac{\pi}{4}\right)$$
$$=V_0\sin\left(\frac{3\pi}{2}+t+\frac{\pi}{4}\right)$$
$$i=\frac{V_0}{2}\sin\left(t+\frac{3\pi}{2}+\frac{\pi}{4}-\phi\right)=\frac{V_0}{\sqrt{2}}\sin\left(\frac{3\pi}{2}+t\right)$$
$$i=-\frac{V_0}{\sqrt{2}}\cos t\Rightarrow V_R=iR=-\frac{V_0}{\sqrt{2}}\cos t$$

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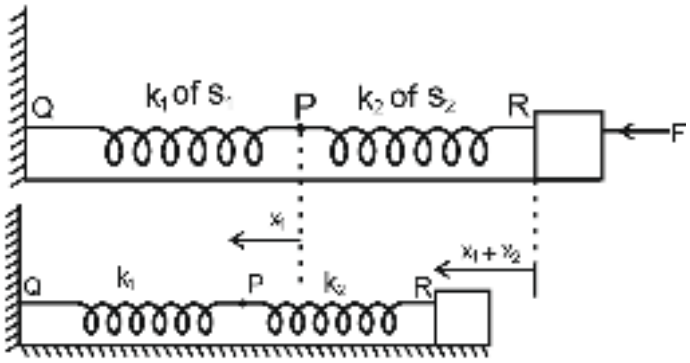
7)

$$B_P = \frac{\mu_0 i}{2\pi r} \qquad i = i_d = \epsilon_0 A \frac{dE}{dt}$$

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0} \Rightarrow \quad S = \frac{EB}{\mu_0}$$

$$S = \frac{E}{\mu_0} \times \frac{H_0 i}{2\pi r} = \frac{E}{2\pi r} \epsilon_0 \pi r^2 \frac{dE}{dt}$$

$$= \frac{1}{2} \epsilon_0 r E \frac{dE}{dt}$$



8) x_1 is compression of S_1

x_2 is compression of S_2

x is total compression

In series $x_1 + x_2 = x$

$$k_1 x_1 = k_2 x_2 = k_{eq.} x = \left(\frac{k_1 k_2}{k_1 + k_2}\right) x$$

$$x_1 = \left(\frac{k_2}{k_1 + k_2}\right) x, \; x_2 = \left(\frac{k_1}{k_1 + k_2}\right) x$$

1. Work done by Springs, on wall is Zero as point of application Q does not move.
2. Work energy theorem for block

$$W_F + W_{spring} = 0$$

$$W_F = -W_{spring} = \frac{1}{2} K_{eq.} x^2$$

as $W_{spring} = U_i - U_f$

$$= 0 - \frac{1}{2} K_{eq.} x^2$$

$$W_F = \frac{1}{2} \left(\frac{k_1 k_2}{k_1 + k_2}\right) x^2$$

3. Work done by S_2 on block k is = – W_F

OR

$$dW_{S2} = \vec{F}_{Sp_2} \cdot \vec{dS} = -k_2 x_2 dx$$

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$$= -k_2\left(\frac{k_1 x}{k_1+k_2}\right)dx$$

$$\int_0^x dW = -\frac{1}{2}\left(\frac{k_1 k_2}{k_1+k_2}\right)x^2$$

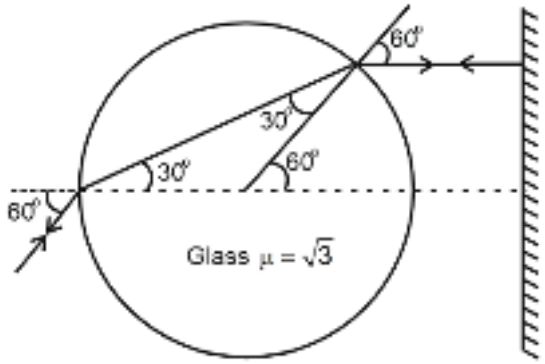
4. Work done by S_2 on spring S_1 is

$$dW_{s_2/s_1} = \vec{F}_{sp_2}.\vec{ds} = k_2 x_2 dx_1$$

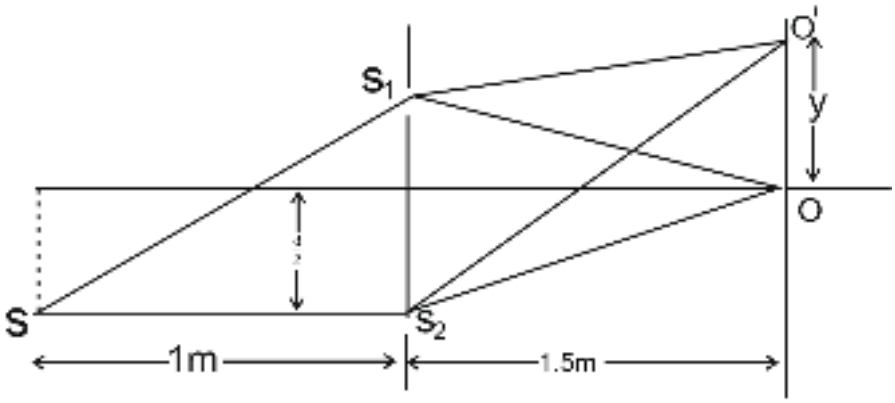
$$\Rightarrow \int k_1 x_1 dx_1 = \frac{1}{2}K_1 x_1^2$$

$$= \frac{1}{2}k_1\left(\frac{k_2}{k_1+k_2}x\right)^2 = \frac{1}{2}\frac{k_1 k_2^2}{(k_1+k_2)^2}x^2$$

9)



10)



At O' Central maxima occured

$$SS_1 + S_1O' = SS_2 + S_2O'$$

$$SS_1 - SS_2 = S_2O' - S_1O'$$

$$\frac{d/2\,d}{1} = \frac{yd}{3/2} \Rightarrow y = \frac{3}{4}d = 0.75\,mm$$

At $O \Rightarrow \Delta x$ = Path Difference $= (SS_1 + S_1O) - (SS_2 + S_2O)$

$$= (SS_1 - SS_2) = \frac{(d/2)\,d}{D}$$

$$= 5000\,\mathring{A} = \lambda$$

11)

here T.E is total energy of electron

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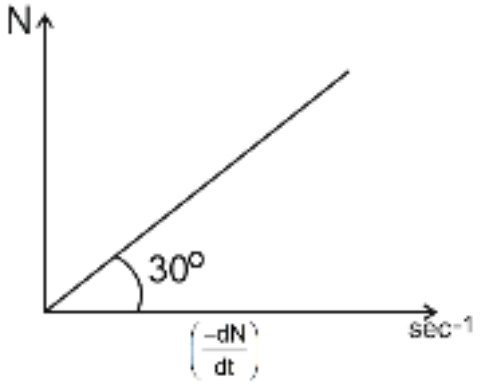
Energy level diagram of hydrogen like atom is

$$\{(T.E)_n - (T.E)_{n-1}\} > \{(T.E)_{n+1} - (T.E)_n\}$$

$$(T.E)_n > \frac{T.E_{n+1} + T.E_{n-1}}{2}$$

$$-E_n > \frac{-E_{n+1} - E_{n-1}}{2}$$

$$E_n < \left(\frac{E_{n+1} + E_{n-1}}{2}\right)$$



12)

Slope of graph $= \frac{N}{\left(\frac{-dN}{dt}\right)} = \frac{1}{\lambda} = \tan 30^\circ$

Law of Radio-active disintegration $= \frac{-dN}{dt} = \lambda N$

$$\lambda = \sqrt{3}\frac{\ell n 2}{T_{1/2}}$$

$$T_{1/2} = \frac{\ell n 2}{\sqrt{3}}$$

13)

Process $A \to B$ is Isothermal

$$PV = nRT \Rightarrow PV = \frac{m}{M}RT$$

$$P = \frac{\rho RT}{M}$$ At constant temperature

$$P \alpha \rho$$

$$\Delta T = 0, \quad \Delta U = 0$$

$$\Delta Q = \Delta W$$

$$\Delta Q_{A\to B} = nRT_A \ell n \frac{V_B}{V_A} = nRT_A \ell n \frac{P_A}{P_B}$$

$$= -nRT_A \ell n 2 = -P_A V_A \ell n 2$$

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$= -P_0 \frac{M}{\rho_0} \ell n 2 = -\frac{P_0 M \ell n 2}{\rho_0}$

$= \frac{P_0 M \ell n 2}{\rho_0}$

Heat rejected

14)

If $\vec{F}_{external} = 0$

V_{cm} = Constant

There will not be any change in velocity of centre of mass of system hence Statement I is wrong

K.E. of system in ground frame = $K_1 + K_2$

where K_1 = K.E. of C.O.M with respect to ground frame

$= \frac{1}{2} m_{system} V_{com}^2$

K_2 = K.E. of system with respect to C.O.M of system

for two particle system

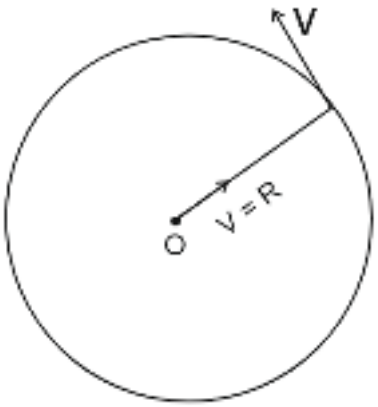
$K_2 = \frac{1}{2}\mu \left|\vec{V}_{rel}\right|^2$

μ = reduced mass

$= \left(\frac{m_1 m_2}{m_1 + m_2}\right)$

$\vec{V}_{rel} = \vec{V}_1 - \vec{V}_2$

Under no external force mechanical energy remains conserved



15)

$F_{cen}|_{r=R} = -\frac{2}{R^2}$

$\left.\frac{mv^2}{r}\right|_{r=R} = \frac{2}{R^2}$

$K.E. = \frac{1}{R}$

$F = -\frac{dU}{dr} = -\frac{2}{r^2} \Rightarrow dU = \frac{2}{r^2} dr$

$U_r = -\frac{2}{r} \Rightarrow U_{r=R} = -\frac{2}{R}$

$T.E = K.E + P.E = \frac{1}{R} - \frac{2}{R} = -\frac{1}{R}$

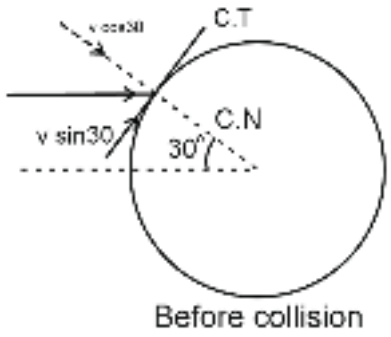
OR

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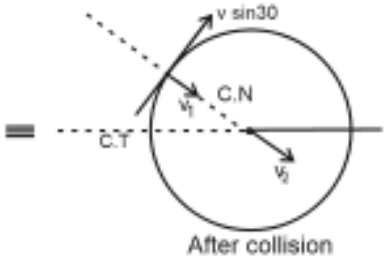
for Inverse square law force

If $F_{cen} \alpha \frac{1}{r^2} \Rightarrow T.E = -KE = -\frac{1}{R}$

16)



Before collision



After collision

Velocity of particle along common Tangent before & after collision remains unchanged $= \frac{v}{2}$

Apply Principle of conservation of linear momentum along Common Normal

$m_1u_{1n} + m_2u_{2n} = m_1v_{1n} + m_2v_{2n}$

$mv \cos 30^\circ + 0 = + mv_1 + mv_2$

$v_2 + v_1 = \frac{v\sqrt{3}}{2}$ _____(1)

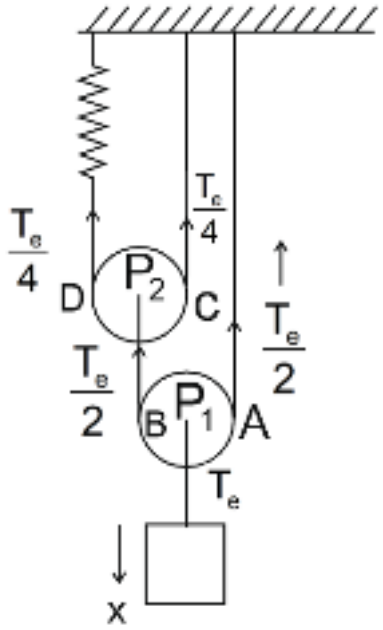
$v_{2n} - v_{1n} = e(u_{1n} - u_{2n})$

$v_2 - v_1 = 0.5\ v \cos 30^\circ = \frac{u\sqrt{3}}{4}$ ___(2)

$v_1 = \frac{v\sqrt{3}}{8}$ speed of particle after Collision = v

$\sqrt{v_{tan}^2 + v_N^2} = \sqrt{\frac{v^2}{4} + \frac{3v^2}{64}} = \frac{\sqrt{19}v}{8}$

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17)

- If System is in Equilibrium Extra Tension T_e in String After Displacing block x from Equilibrium

$x_B = 2x = x_{P2}$

$x_D = 4x$

$\frac{T_e}{4} = K\,x_D = 4Kx$

$T_e = 16Kx \Rightarrow T_e = -16Kx$

$$T_e = 2\pi\sqrt{\frac{m}{16K}}$$

Ans (A)

II Method

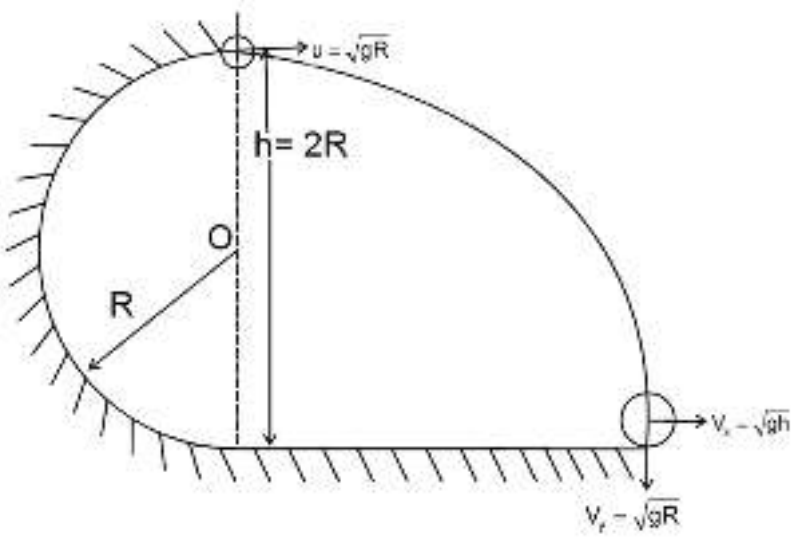
If gravity neglected

$$U_{spring} = \frac{(F_{spring})^2}{2k} = \frac{\left(\frac{T_e}{4}\right)^2}{2k} = \frac{T_e^2}{32k}$$

$$\frac{T_e^2}{2K_{eff.}} \Rightarrow k_{eff.} = 16K$$

Time period = $2\pi\sqrt{\frac{m}{k_{eff.}}} = 2\pi\sqrt{\frac{m}{16K}}$

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18)

v_x = horizontal velocity just before Ist collision = $\sqrt{gR}$

v_x'= horizontal velocity just after Ist collision = $\sqrt{gR}$

v_y = vertical velocity just before Ist collision = $\sqrt{2gh} = \sqrt{2g \times 2R} = 2\sqrt{gr}$

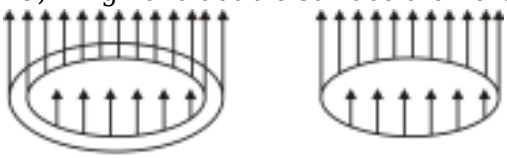
v_y'= vertical velocity just after Ist collision

$v_y' = ev_y = 0.5 \times 2\sqrt{gR} = \sqrt{gR}$

$$R = \frac{2v_x'v_y'}{y}$$

$$= \frac{2 \times \sqrt{gR} \times \sqrt{gR}}{g} = 2R$$

19) Ring have double surface than that of disk.



20) By Bernoulli's theorem

$$P + \frac{1}{2}\rho v^2 + \rho gh = C$$

h is same for all point

so $P + \frac{1}{2}\rho v^2 = C$

so if V is more than P is less.

for A & C cross section area is same so V is also same so P also same so height of liquid will be same for A & C.

21)

Viscous force = $F_v = \eta A\left(\frac{dv}{dy}\right)$

Shear stress = $\frac{F_v}{A} = \eta\left(\frac{dv}{dy}\right)$

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$$v = \alpha\left[\frac{y}{h} - \left(\frac{y}{h}\right)^2\right]$$

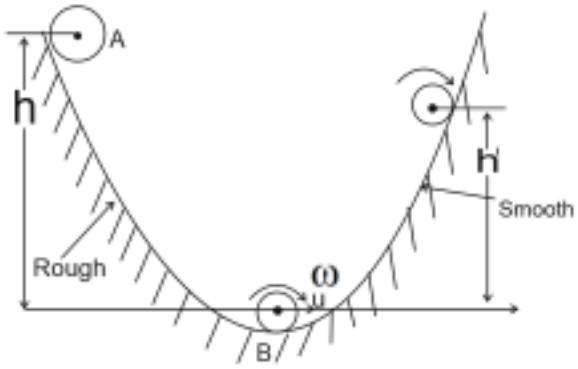
$$\frac{dv}{dy} = \alpha\left[\frac{1}{h} - \frac{2y}{h^2}\right]$$

$$\Rightarrow \left.\frac{dv}{dy}\right|_{y=0} = \alpha\left[\frac{1}{h} - \frac{2x0}{h}\right]$$

$$= \frac{\alpha}{h}$$

shear stress at baseplate $= \eta\frac{\alpha}{h}$

N = 1



22)

Total energy at bottom = mgh

$$K.E_{rot} = \frac{1}{2}I_c\omega^2$$

$$K.E_{trans} = \frac{1}{2}mu^2$$

$$\frac{K.E_{Rot}}{K.E_{trans}} = \frac{2}{5}$$

$$K.E_{Rot} = \frac{2mgh}{7}, \quad K.E_{trans} = \frac{5mgh}{7}$$

At top most point translation k.E. is converted into grav. potential energy

$$mgh' = \frac{5}{7}mgh$$

$$\Rightarrow h' = \frac{5}{7}h$$

h'= 5m



23)

$$T(x) = \frac{Mgx}{L}$$

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$$U(x) = \frac{1}{2}\int_{2}^{l} \frac{T^2(x)}{A^2Y} A dx$$

$$U(x) = \frac{M^2g^2}{2L^2AY}\int_{\frac{l}{2}}^{l} x^2 dx$$

$$= \frac{M^2g^2}{2AL^2Y} \times \frac{1}{3}\left[\frac{7l^3}{8}\right]$$

24) $\eta = \frac{1}{3} = \frac{W_{net}}{Q_S}$

$\frac{1}{3} = \frac{Q_S - Q_R}{Q_S} \Rightarrow 1 - \frac{Q_R}{Q_S} = \frac{1}{3}$

$\frac{Q_R}{Q_S} = \frac{2}{3}$

$Q_R = 1 \times 80 \times 4.2$

$Q_R = 336000$

$W + Q_R = Q_S$

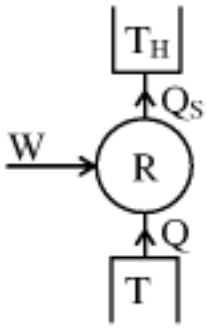
$\because \frac{Q_R}{Q_S} = \frac{2}{3}$

$Q_S = \frac{3}{2}Q_R$

$W + Q_R = \frac{3}{2}Q_R$

$W = \frac{1}{2}Q_R$

$W = \frac{1}{2}Q_R \Rightarrow \frac{1}{2}(336000)$

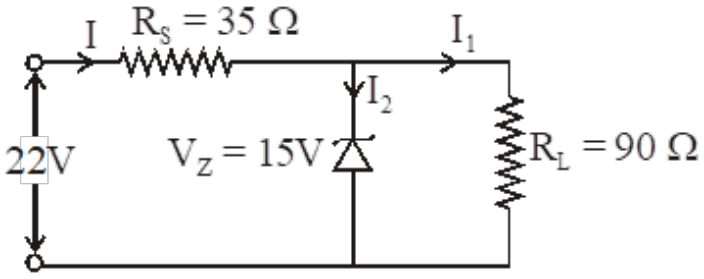


$P \times t = W$

$(186)t = 168000$

$t = 903.2$ sec

$= 15.05$ min



25)

Voltage across RS = 22 – 15 = 7V

Current through RS = $I = \frac{7}{35} = \frac{1}{5}A$

Current through 90Ω = $I_2 = \frac{15}{90} = \frac{1}{6}A$

Current through zener = $\frac{1}{5} - \frac{1}{6} = \frac{1}{30}A$

Power through zener diode

P = VI

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$P = 15 \times \frac{1}{30} = 0.5$ watt

$P = 5 \times 10^{-1}$ watt

PART-2 : CHEMISTRY

26)

1st line : $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$\therefore \lambda = 660$ nm $\Rightarrow$ Red

27)

$Be(g) \rightarrow Be^-(g)$ (endothermic)

$Be^-(g) \rightarrow Be(g)$ (exothermic)

$P(g) \rightarrow P^{3-}(g)$ (endothermic)

$H^+(g) \rightarrow H(g)$ (exothermic)

$H(g) \rightarrow H^-(g)$ (exothermic)

$O^-(g) \rightarrow O(g)$ (endothermic)

$O(g) \rightarrow O^+(g)$ (endothermic)

28) $\underset{(aq)}{Pb(OH)_2} + \underset{(aq)}{NaOH} \rightarrow \underset{(aq)}{Na_2PbO_2} + H_2O$

$Pb(OH)_2$ is an amphoteric hydroxide.

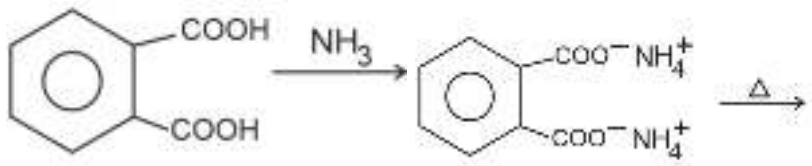
29) $N \equiv N$ has highest energy

According to Molecular orbital theory, between C_2 and O_2, O_2 has more antibonding electron than C_2

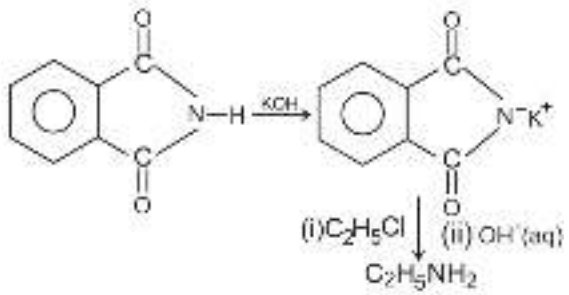
bond energy of O_2 is less than bond energy of C_2.

30) $\Delta T_f = iK_f m$

lower the ΔT_f, higher the freezing point.

31) 

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33)

Let the mass of organic compound taken = m g
Volume of H_2SO_4 of molarity, M.
taken = V mL
Volume of NaOH of molarity, M, used for titration of excess of H_2SO_4 = V_1 mL
V_1 mL of NaOH of molarity M
= $V_1/2$ mL of H_2SO_4 of molarity M
Volume of H_2SO_4 of molarity M unused
= $(V - V_1/2)$ mL
$(V - V_1/2)$ mL of H_2SO_4 of molarity M
= 2 $(V - V_1/2)$ mL of NH_3 solution of molarity M.
1000 mL of 1 M NH_3 solution contains 17 g NH_3 or 14 g of N
$2(V - V_1/2)$ mL of NH_3 solution of molarity M contains:

$$\frac{14 \times M \times 2(V - V_1/2)}{1000} \text{ g N}$$

Percentage of $N = \frac{14 \times M \times 2(V - V_1/2)}{1000} \times \frac{100}{m}$

$$N = \frac{1.4 \times M \times 2(V - V_1/2)}{m}$$

NCERT

34)

NCERT:
$10^9 \rightarrow$ giga,
$10^{21} \rightarrow$ zeta,
$10^{-18} \rightarrow$ atto,
$10^{18} \rightarrow$ exa

35) $\frac{R_1}{R_2} = \frac{R_3}{R_4}$

$$\frac{100}{100} = \frac{25}{R_4}$$

$R_4 = 25\Omega$

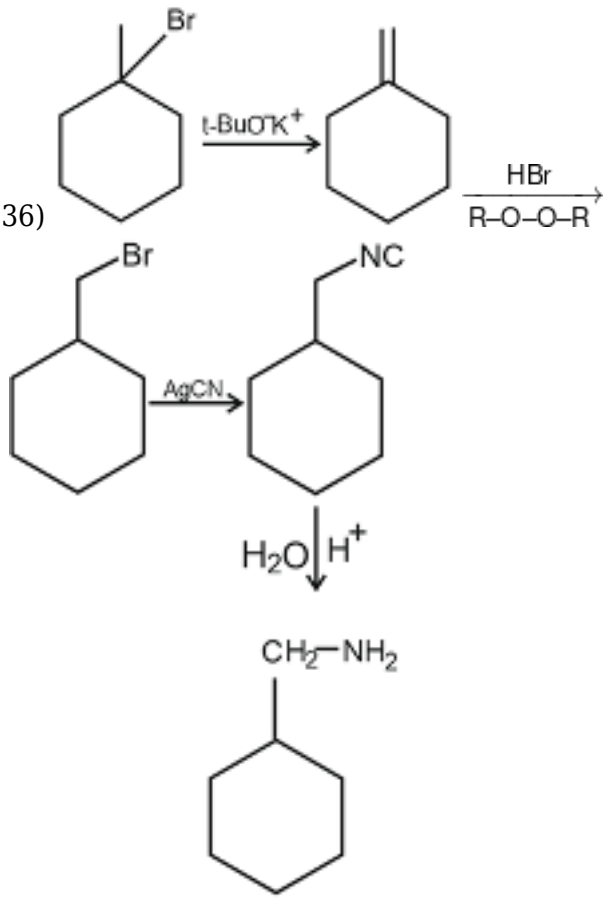
$$\lambda_m = \frac{\kappa \times 1000}{M}$$

$$\lambda_m = \frac{1}{R} \times G^* \times \frac{1000}{M}$$

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$$\lambda_m = \frac{1.25}{25} \times \frac{1000}{0.05}$$
$$\lambda_m = 1000\ Scm^2mol^{-1}$$

36)



37)

NCERT

38)

For,
n = 0, $[Ni(NH_3)_6]^{2+}$ no. of stereo isomers = 0
n = 1, $[Ni(NH_3)_4(en)]^{2+}$ no. of stereo isomers = 0
n = 0, $[Ni(NH_3)_2(en)_2]^{2+}$ no. of stereo isomers = 3
n = 0, $[Ni(en)_3]^{2+}$ no. of stereo isomers = 2

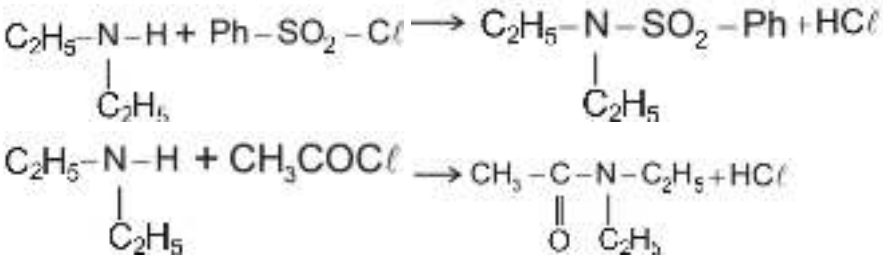
39) $C_2O_4^{2-}$ act as strong field ligand with Co^{+3}.

40) $Ph - CHO$ will give positive tollen's test and negative fehling's test.
Ph – CH_2 – CHO will give positive fehling's test.

41)

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2° Amine give reaction Hinsberg reagent $(Ph-SO_2-C\ell)$ and 2° Amine does not give carbylamine test . (reaction with KOH and $CHC\ell_3$)



→ Amine does not reacts with HNO_2
→ Amine does not give mustard oil test.

42)

NCERT

43) $2A(g) \rightarrow B(g)$
$P = P_0 - P_T$
rate of reaction $\Rightarrow \frac{-1}{2}\frac{d[A]}{dt} = r$
also,
r = k[A]
$\Rightarrow \frac{-1}{2}\frac{d[A]}{dt} = K[A]$
$\Rightarrow P_t = P_0 - 2P$
$\Rightarrow 2P_T - P_0$
$\Rightarrow k = \frac{1}{2t}\ell n\frac{P_0}{2P_T - P_0}$

44) $3Cg + 4H_2 \rightarrow C_3H_8$
$\Delta_f H(C_3H_8) =$
$-104 = 3 \times \Delta_C H^{o}(C) + 4 \times \Delta_C H^{o}(H_2) - \Delta_C H^{o}(C_3H_8)$
$-104 = -1179 + (-1140) - \Delta_C H^{o}(C_3H_8)$
$\Delta_C H^{o}(C_3H_8) = -2215\ kJmol^{-1}$
$\Delta_{reaction} H^{o} = \Delta_C H^{o}(C_3H_8) + \Delta_C H^{o}(H_2) - [\Delta_C H^{o}(C_2H_6) + \Delta_C H^{o}(CH_4)]$
$\Delta Hrxn \Rightarrow -2215 + (-285) - [-1560 + (-890)]$
$\Rightarrow -2500 - [-2450]$
$\Rightarrow -2500 + 2450$
$\Rightarrow -50\ kJmol^{-1}$

45)

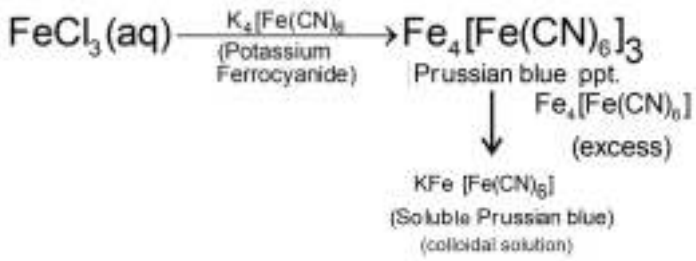
NCERT

46)

single super pair = multiple Bond

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NCERT.



47)

48)

Mass of Tripeptide
= molar mases of (Glycine + Alanine + Leucine) – 2 × molar mass of water
= 75 + 89 + 131 – 36
= 259

49)

2° Alcohol give turbidity in 5-10 minutes with Lucas reagent.
(Isopropyl alcohol, sec-Butyl alcohol)

50)

No change in state function.

PART-3 : MATHEMATICS

51) $s = 2 + \frac{5}{7} + \frac{8}{49} + \frac{11}{343} + ...\infty...(1)$

$s.\frac{1}{7} = \frac{2}{7} + \frac{5}{7^2} + \frac{8}{7^2} + ...\infty...(2)$

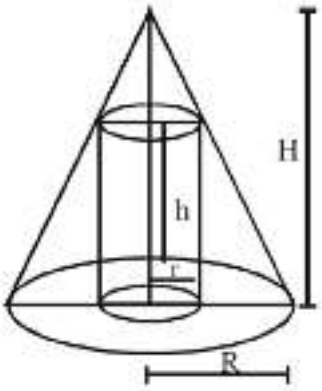
(1) – (2)

$S - \frac{S}{7} = 2 + \frac{3}{7} + \frac{3}{7^2} + \frac{3}{7^3} + ...$

$\Rightarrow \frac{6S}{7} = 2 + \frac{3}{6} \Rightarrow \frac{6S}{7} = \frac{5}{2}$

$\Rightarrow S = \frac{35}{12}$

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52)

$\because \frac{H}{R} = \frac{H-h}{r} \Rightarrow h = \frac{H}{R}(R-r)....(1)$

$\therefore V_2 = \pi r^2 h = \pi r^2 \frac{H}{R}(R-r)$

$\Rightarrow V_2 = \frac{\pi H}{R}(Rr^2 - r^3)$

put $\frac{dV_2}{dr} = 0 \Rightarrow r = \frac{2R}{3}, h = \frac{H}{3}$

$\therefore \frac{V_1}{V_2} = \frac{\frac{1}{3}\pi R^2 H}{\pi . \frac{4R^2}{9} . \frac{H}{3}} = \frac{9}{4}$

53) Reason : Let a = 2□ + 1, b = 2m + 1 & c = 2n + 1 where □,m,n ∈ I

Equation $ax^2 + bx + c = 0$ will have rational roots if $b^2 - 4ac$ is a perfect square.

Let $b^2 - 4ac$ is a perfect square

$\Rightarrow (2m + 1)^2 - 4(2□ + 1)(2n + 1) = (2k + 1)^2$

(let), k ∈ I

or $(2m + 1)^2 - (2k + 1)^2 = 4(2□ + 1)(2n + 1)$

$\Rightarrow 2(m - k)\, 2(m + k + 1) = 4(2□ + 1)(2n + 1)$

RHS is product of 2 odd integers.

□ LHS should also contain product of 2 odd integers.

⇒ m – k & m + k + 1 has to be odd, which is not possible for any values of m,k ∈ I.

Hence Reason is true.

Assertion : Line 7x + 5y + 1 = 0 will intersect the parabola $y = x^2$ in rational point if equation $5x^2 + 7x + 1 = 0$ has rational roots.

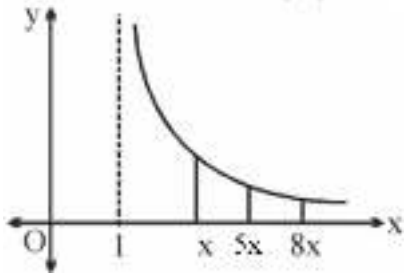
Hence using Reason, Assertion is true.

54) As f(x) is a continuous decreasing positive function on (1, ∞), so

$0 < f(8x) < f(5x) < f(x)$

$\Rightarrow 1 < \frac{f(5x)}{f(8x)} < \frac{f(x)}{f(8x)}$

$\Rightarrow \lim_{x\to\infty} 1 \leqslant \lim_{x\to\infty} \frac{f(5x)}{f(8x)} \leqslant \lim_{x\to\infty} \frac{f(x)}{f(8x)}$

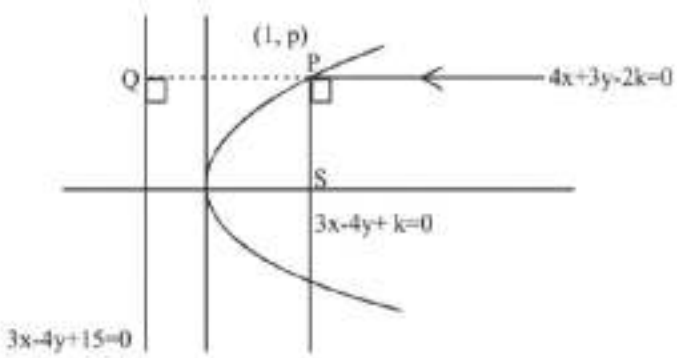


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 By squeeze play theorem,

$$\lim_{x\to\infty}\frac{f(5x)}{f(8x)} = 1$$

55)



(1, p) satisfies

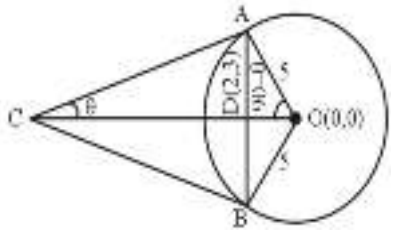
$4x + 3y - 2k = 0$ & $3x - 4y + k = 0$

 $p = 2$ & $k = 5$

now p = (1,2)

Now PS = PQ = 2

 LR = 4

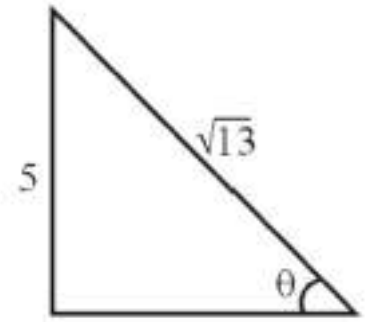


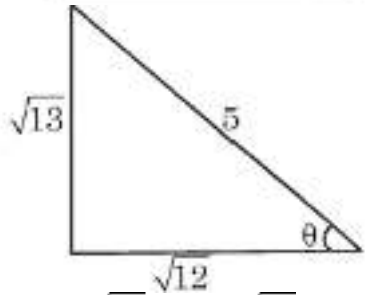
56)

Let D is the mid point of ABD (2, 3)

OD = $\sqrt{3}$

In ΔAOD, $\cos(90^\circ - \theta) = \frac{\sqrt{13}}{5}$





$\sin\theta = \frac{\sqrt{13}}{5}, \tan\theta\frac{\sqrt{13}}{\sqrt{12}}$

Now ln ΔCAO,

$\tan\theta\frac{5}{CA}$

$\frac{5}{CA} = \frac{\sqrt{13}}{\sqrt{12}} \Longrightarrow \frac{5\times\sqrt{12}}{\sqrt{13}} = CA$

Now area of OACB = CA × 5

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$= \frac{5\sqrt{12} \times 5}{\sqrt{13}}$

$50\sqrt{\frac{3}{13}}$ sq. unit

$= 141$

57) The required no of ways

$= {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + ...+{}^{10}C_6 + {}^{10}C_7$

$= 2^{10} - ({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10})$

$= 2^{10} - 2(1 + 10 + 45)$

$= 912$

58) $\frac{a-1}{2} = \frac{5-a}{3} \Rightarrow a = \frac{13}{5}$

59) (A) $1 - 8(\log_{10}x)^2 = \log_{10}x - 2(\log_{10}x)^2 \Rightarrow 6(\log_{10}x)^2 + \log_{10}x - 1 = 0$

$\log_{10}x = \frac{1}{3}, -\frac{1}{2}$

$z = 10^{1/3}, 10^{-1/2}$

$\alpha = 10^{-1/2}, \beta = 10^{1/3}$

$\frac{(\beta^3 + 1)}{\alpha^2\beta^3} = \frac{(10 + 1)}{10^{-1}10} = 11$

(B) For any $x > 0$, L.H.S. > 0 and R.H.S. $< O$

No value of x will satisfy the given equation. Hence no solution.

(C) $|x| + \left|\frac{4-x^2}{x}\right| = \left|x + \frac{4-x^2}{x}\right|$

as $|a| + |b| = |a + b| \Rightarrow ab \geqslant 0$

$x\left(\frac{4-x^2}{x}\right) \geqslant 0, x \neq 0$

$\Rightarrow x^2 - 4 \leq \Rightarrow x \in [-2,2] - \{0\}$

(D) $\log_b 125 = \frac{\log_c 8}{\log_c 2} \Rightarrow \log_b 125 = 3$

$\Rightarrow b^3 = 125 \Rightarrow b = 5$

60) Since A.M. of n observations is M.

 Total of n observations = Mn

sum of (n – 4) observations = a

 Sum of remaining observations = Mn – a

 Required Mean = $= \frac{Mn - a}{4}$

61) $I = \int \frac{dx}{(\sqrt{x})^2 + (\sqrt{x})^7} = \int \frac{dx}{x^{7/2}(1 + x^{-5/2})}$

Now Put $x^{-5/2} = t \Rightarrow \frac{-5}{2x^{7/2}}dx = dt$

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$$\Rightarrow I = \frac{-2}{5}\int \frac{dt}{1+t} = -\frac{2}{5}\ell n|1+t| + c$$

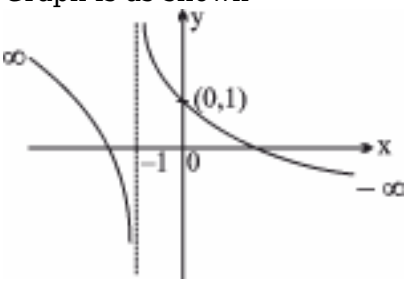
$$\Rightarrow \frac{2}{5}\ell n\left|\frac{1}{1+t}\right| + c \Rightarrow \frac{2}{5}\ell n\left|\frac{x^{5/2}}{1+x^{5/2}}\right| + c$$

62) $f'(x) = -\frac{3}{(x+1)^4} - 3 + \cos x < 0$

hence f (x) is always decreasing, Also as $x \to \infty$, $f(x) \to -\infty$ and as $x \to -\infty$, $f(x) \to +\infty$

hence one positive and one negative root

Graph is as shown



63) n^{th} term of 3, 7, 13, 21, 31 is $n^2 + n + 1$ $T_n = \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\frac{(n+1)-n}{1+n(n+1)} =$ $\tan^{-1}(n+1) - \tan^{-1} n$

Hence

$S = T_1 + T_2 + T_3 + T_4 + T_5$

$= (\tan^{-1}2 - \tan^{-1}1)$

$(\tan^{-1}3 - \tan^{-1}2)$

$(\tan^{-1}4 - \tan^{-1}3)$

$(\tan^{-1}5 - \tan^{-1}4)$

$(\tan^{-1}6 - \tan^{-1}5)$

$S = \tan^{-1}6 - \tan^{-1}1 = \tan^{-1}\left(\frac{6-1}{1+6}\right) = \tan^{-1}\left(\frac{5}{7}\right)$

$\therefore \tan\left(\tan^{-1}\frac{5}{7}\right) = \frac{5}{7}$

integer $= 14 \cdot \frac{5}{7} = 10$

64) $L = \lim_{x\to 0} -\left(\frac{1-\cos x}{x^2}\right)\left(\frac{(e^{2x^2}-1-2x^2)x^2}{x^2 \cdot x^2(\sin 3x - \ln(1+3x))}\right)$

$= -\frac{1}{2}\lim_{t\to 0}\frac{e^{2t}-1-2t}{t^2} \cdot \lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$ where $x^2 = t$

$= -\frac{1}{2}\lim_{y\to 0}\frac{4(e^y-1-y)}{y^2} \cdot \lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$

$= -\frac{1}{2}.2.\lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$

now, $L = \lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$

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using L'Hospital's rule $L = \frac{2}{9}$;

$\therefore \quad L = -\frac{2}{9}$

65) f ″ (x) is changing its sign at $x = 1,\ 2,\ 3,\ 5$
$\Rightarrow$ 4 points of inflexion
$\Rightarrow S_2$ is also true
Whereas $x = 4,\ 6$ are points of
extrema for $f(x),\ [f'(4) = f'(6) = 0]$
$\Rightarrow S_1$ is false
$\therefore$ C is correct

66) If the ellipse & hyperbola are orthogonal
$\Rightarrow$ they are confocal
For ellipse : $e^2 = 1 - \frac{7}{16}$
$\therefore e = \frac{3}{4}$
$\Rightarrow ae = 4 \times \frac{3}{4} = 3$
For Hyperbola :-
$\frac{x^2}{144} - \frac{y^2}{\alpha} = \frac{1}{25}$
$\frac{x^2}{\left(\frac{144}{25}\right)} - \frac{y^2}{\left(\frac{\alpha}{25}\right)} = 1$
$e_1^2 = 1 + \frac{\alpha}{144}$
$e_1 = \sqrt{1 + \frac{\alpha}{144}}$
$ae = a_1 e_1$ (focus of both coincide)
$\Rightarrow 3 = \frac{12}{5} \cdot \sqrt{1 + \frac{\alpha}{144}}$
$\Rightarrow \frac{9 \times 25}{144} = 1 + \frac{\alpha}{144}$
$\Rightarrow \alpha = 81$
$\therefore e_1^2 = 1 + \frac{81}{144}$
$\Rightarrow e_1 = \frac{15}{12} = \frac{5}{4}$

67) $gof(x) = x$
$g'(f(x)) = \frac{1}{f'(x)}$
$g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$
At $f(x) = 2,\ 2^x + x^3 + 1 = 2 \Rightarrow x = 0$

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$f'(x) = 2^x \ln 2 - 3x^2 + 0$

$F'(0) = \ln 2$

$F''(x) = 2^x (\ln 2)^2 + 6x$

$F''(0) = (\ln 2)^2$

$g'(2) = \dfrac{1}{f'(0)} = \dfrac{1}{\ln 2} = \log_2 e$

$g''(2) = \dfrac{-f''(0)}{(f'(0))^3} = \dfrac{-(\ln 2)^2}{(\ln 2)^3} = -\dfrac{1}{\ln 2} = -\log_2 e$

$\therefore \dfrac{g'(2)}{g''(2)} = -1$

68) Given limit can be written as $\lim\limits_{t\to 0} \dfrac{\left(3^t-1\right)\left(4^t-1\right)\left(\sqrt{2\cos t+7}+3\right)}{2\cos t-2}$

$= \lim\limits_{t\to 0} \dfrac{\left(3^t-1\right)}{t} \dfrac{\left(4^t-1\right)}{t} \dfrac{\left(\sqrt{2\cos t+7}+3\right)}{-2\frac{(1-\cos t)}{t^2}t^2}$

$= \log_e 3 \cdot \log_e 4 \,(-6) = -6\log_e 3.\log_e 4$

69) Let $\vec{a} \wedge \vec{b} = \theta$ Now $\vec{x} = \vec{a} - \left(\vec{a}.\vec{b}\right)\vec{b}$

$\Rightarrow |\vec{x}|^2 = \vec{a}^2 + \left(\vec{a}.\vec{b}\right)^2 b^2 - 2\left(\vec{a}.\vec{b}\right)^2$

$\Rightarrow |\vec{x}|^2 = 1 + \cos^2\theta - 2\cos^2\theta$

$\Rightarrow |\vec{x}| = \sin\theta$

Also $\vec{y} = \vec{a} \times \vec{b} \Rightarrow |\vec{y}| = \left|\vec{a} \times \vec{b}\right| = \sin\theta$

$\therefore |\vec{x}| = |\vec{y}|$

Also, $\vec{x}.\vec{b} = \vec{a}.\vec{b} - \left(\vec{a}.\vec{b}\right)\vec{b}^2 = 0$

$\therefore \left|\vec{x}.\vec{b}\right| = 0$

70) $I_1 = \dfrac{b-a}{2} = \dfrac{\pi/2-0}{2}$ $I_1 = \dfrac{\pi}{4}$

$I_2 = \int\limits_{-\pi}^{\pi} \dfrac{\cos^2(x)}{1+\pi^x}$

apply King and add

$2I_2 = \int\limits_{-\pi}^{\pi} \cos^2(x)\ .dx = 2\int\limits_{0}^{\pi} \cos^2(x)\ .dx$

$\Rightarrow I_2 = \int\limits_{0}^{\pi} \cos^2 x\, dx = \dfrac{\pi}{2}$

71)

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P(all tails or all heads in one trial) = $\frac{1}{4}$

$\therefore$ p(this happens second time in 3rd trial)

= $^{2}C_{1} \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{4} = \frac{3}{32}$

72)

Put

Y = x + 2

73) $1-\sin^2 x + \frac{\left(\sqrt{3}+1\right)}{2}\sin x - \frac{\sqrt{3}}{4} - 1 = 0$

$\sin^2 x - \frac{\left(\sqrt{3}+1\right)}{2}\sin x + \frac{\sqrt{3}}{4} = 0$

$4\sin^2 x - 2\sqrt{3}\sin x - 2\sin x + \sqrt{3} = 0$

on solving we get

$\sin x = 1/2 ; \frac{\sqrt{3}}{2}$

= (π/6, 5π/6 ; π/3, 2π/3]

74) $x^2 - px + 1258 = 0$

Product of roots

$\Rightarrow 1258 = 2 \times 17 \times 37$

$\Rightarrow \alpha + \beta = p$

For smallest p,

$\alpha = 34, \beta = 37$

$p = 34 + 37$

= 71

75) $\therefore [p\ q\ r] = 0$

$$\Rightarrow \begin{vmatrix} x & 1 & 1 \\ 1 & y & 1 \\ 1 & 1 & z \end{vmatrix} = 0$$

$C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$

$$\Rightarrow \begin{vmatrix} x & 1-x & 1-x \\ 1 & y-1 & 0 \\ 1 & 0 & z-1 \end{vmatrix} = 0$$

$\Rightarrow x(y-1)(z-1) - (1-x)(z-1) + (1-x)(1-y) = 0$

$\Rightarrow \frac{x}{x-1} + \frac{1}{y-1} + \frac{1}{z-1} = 0$

$\Rightarrow 1 + \frac{1}{x-1} + \frac{1}{y-1} + \frac{1}{z-1} = 0$

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$$\Rightarrow \frac{1}{x-1}+\frac{1}{y-1}+\frac{1}{z-1}=-1$$
$$\Rightarrow \frac{1}{1-x}+\frac{1}{1-y}+\frac{1}{1-z}=1$$

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